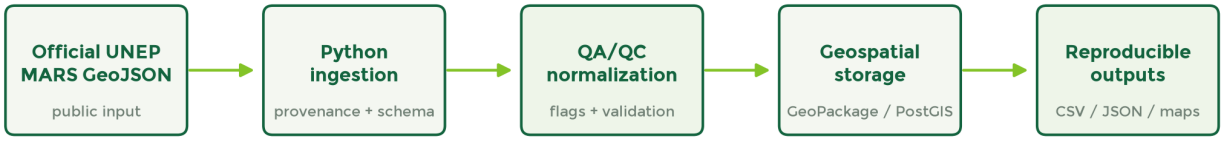
A Reproducible QA/QC Pipeline for Public Satellite Methane Plume Data from UNEP IMEO MARS

A compact geospatial data-engineering case study using the Eye on Methane public dataset

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Technical report	Version 1.0.0
Dataset snapshot	Retrieved 6 October 2026
Report / derived assets	CC BY-NC-SA 4.0
Pipeline source code	MIT License
Zenodo DOI	https://doi.org/10.5281/zenodo.23196446

METHANE publication pipeline



Scope boundary: uses already processed methane detections; no atmospheric retrieval or plume-detection model is implemented.

Independent technical demonstration. This work uses already processed public UNEP IMEO MARS detections. It does not implement atmospheric methane retrieval, radiative transfer, plume-detection models, or an alternative to MARS.

Executive summary

Reliable methane monitoring requires more than retrieval algorithms. Public satellite-derived products also need traceable ingestion, schema normalization, data-quality controls, geospatial persistence, and reproducible outputs before they can support downstream analysis or operational workflows. This report presents a compact pipeline built around the public Methane Alert and Response System (MARS) plume and source data released by UNEP's International Methane Emissions Observatory (IMEO) through Eye on Methane.

30,116	4,186	99	10
plume observations	detected sources	countries	sensor labels

What this demonstration proves

The workflow captures source provenance, ingests GeoJSON, normalizes types, assigns deterministic QA/QC flags, creates a portable geospatial layer, prepares an indexed PostgreSQL/PostGIS loading path, runs automated tests, and exports machine-readable summaries and publication figures.

What it deliberately does not prove

The project does not retrieve XCH4 from radiances, estimate methane enhancements, create plume masks, quantify emissions independently, train machine-learning models, or reproduce UNEP IMEO scientific retrieval methods.

Key QA/QC outcome

Status	Records	Share	Interpretation
PASS	27,871	92.545%	No defined checks triggered
WARN	2,245	7.455%	Only non-critical issues
FAIL	0	0.000%	No critical structural failures

A PASS result here is a downstream structural QA result. It is not an independent scientific validation of the methane retrieval.

1. MARS context and data provenance

UNEP IMEO's Methane Alert and Response System is a public global satellite detection and notification system for very large methane emissions. UNEP describes MARS as integrating data from more than 30 satellite instruments with scientific expertise and analytical methods to support mitigation action. This report focuses on the downstream engineering layer once a validated public product is available.

Official public input

The pipeline uses the Eye on Methane MARS GeoJSON export of detected plumes and sources. The data dictionary defines a plume as a single methane emission observation from one satellite overpass and a source as a ground location where one or more plumes have been detected over time.

Provenance item	Snapshot value
Download date	6 October 2026
Plume records	30,116
Source records	4,186
Observation window	2020-10-29 to 2026-09-05 UTC
Archive bytes	54,940,454
SHA-256	3755cc5206f6c60c770fe589d527a98f73779e16fe752f40dda6e120935ed03e

The official package also includes a data dictionary and the CC BY-NC-SA 4.0 license. The retrieval timestamp, source URL, archive size, file list and cryptographic hash are stored in `data/raw/provenance.json`.

Snapshot sensitivity. The MARS data dictionary states that Eye on Methane plumes detected by Sentinel-2, Landsat-8/9, EMIT, PRISMA and EnMAP were requantified after 20 February 2026. A cryptographic snapshot identifier is therefore essential for reproducibility.

2. Data model and normalization

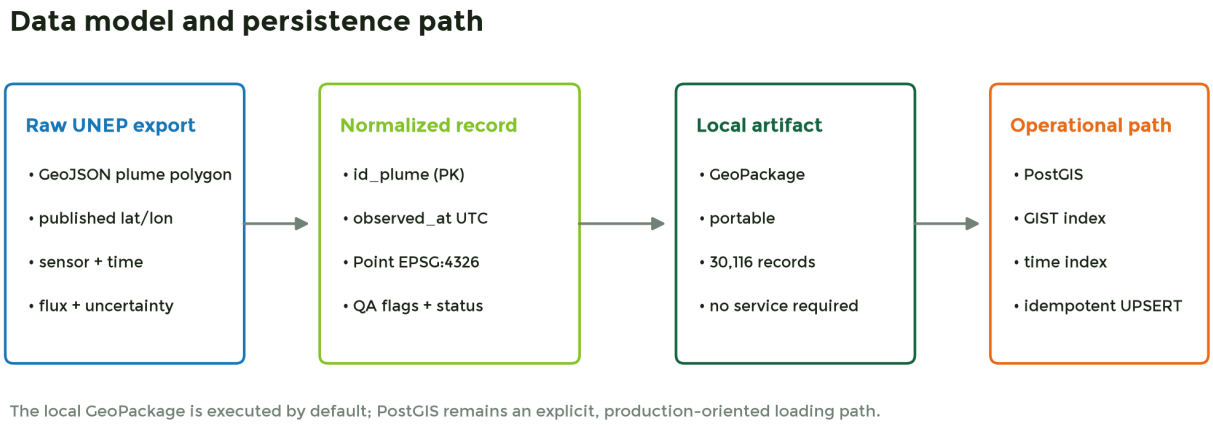


Figure 1. Data model and persistence path used by the demonstration.

The raw plume export contains identifiers, satellite/instrument names, timestamps, WGS84 latitude and longitude, sector attributes, actionability/notification fields, methane flux-rate estimates and uncertainty fields, wind information, and optional original plume geometries.

Normalized field	Purpose	QA behavior
id_plume	Unique record key	Missing or duplicated -> FAIL
observed_at_utc	Canonical UTC timestamp	Invalid/missing -> FAIL
lat / lon	Published WGS84 location	Missing/out of range -> FAIL
geometry	Point EPSG:4326 from valid lat/lon	Built only from valid coordinates
ch4_fluxrate	Published kg/h estimate	Missing -> WARN; negative -> FAIL
has_original_footprint	Preserves upstream geometry availability	Missing footprint -> WARN
qa_status	PASS / WARN / FAIL	Derived from deterministic flags

The normalized persistent geometry is a point derived from the published latitude/longitude fields. The workflow does not reconstruct missing plume polygons. Instead, original footprint availability is retained as an explicit QA field.

3. Reproducible QA/QC

The QA/QC layer is deterministic and record-level. Critical checks cover identity, time, coordinate validity and physically invalid negative flux values. Warning checks cover missing published flux values and missing/invalid upstream plume geometries.

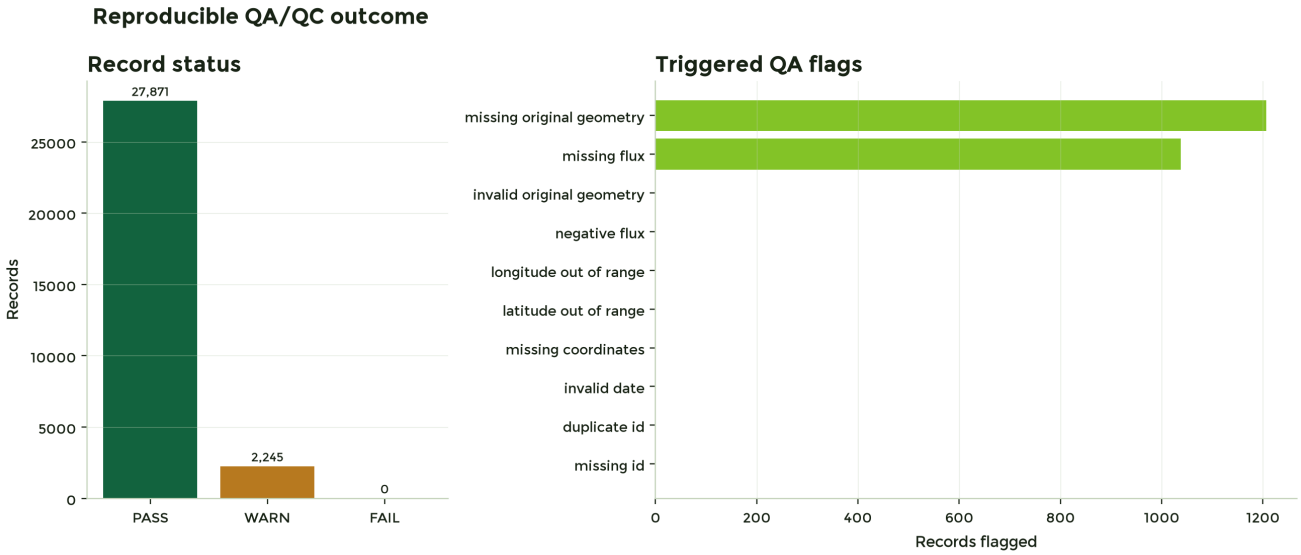


Figure 2. Record-level QA status and triggered validation flags.

Check	Count	Class
Missing CH4 flux rate	1,038	WARN
Missing original geometry	1,207	WARN
Missing/duplicate ID	0	FAIL
Invalid/missing date	0	FAIL
Missing/out-of-range coordinates	0	FAIL
Negative flux rate	0	FAIL
Invalid original geometry	0	WARN

A missing original footprint is treated as a warning rather than a failure when valid latitude/longitude coordinates remain available for downstream location-level ingestion.

4. Multi-sensor snapshot composition

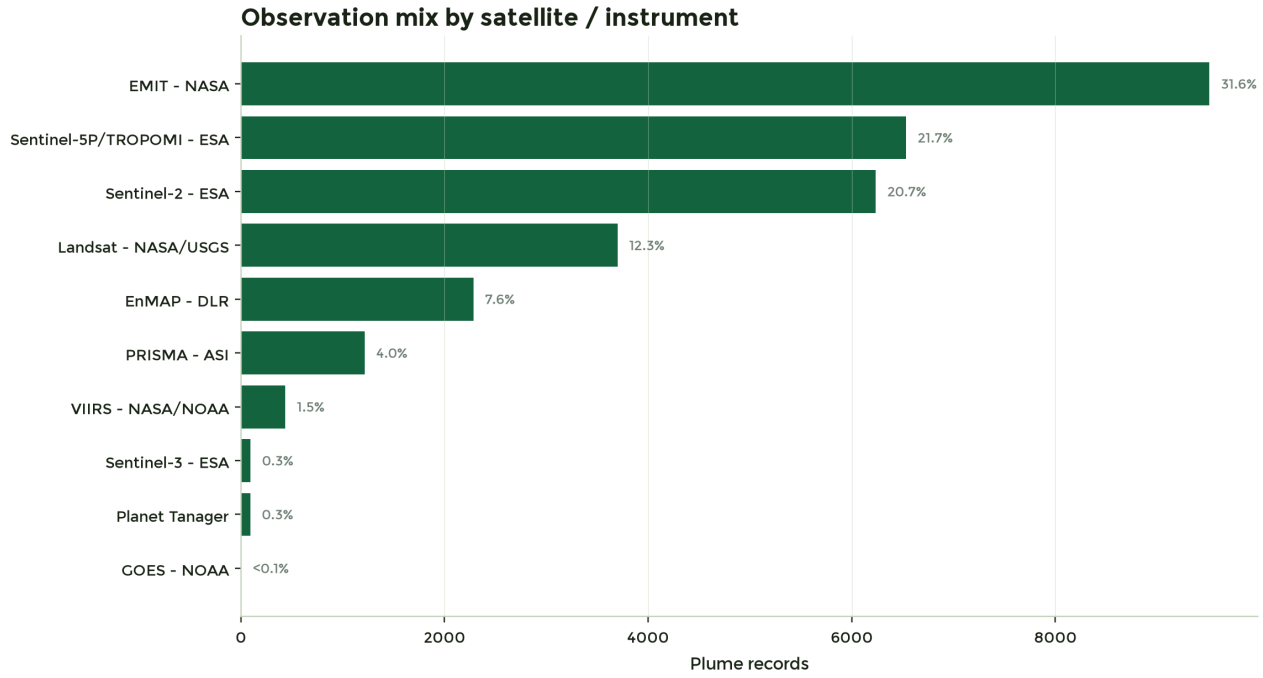


Figure 3. Distribution of plume observations by satellite/instrument label in the downloaded snapshot.

EMIT accounts for 31.6% of plume records, followed by Sentinel-5P/TROPOMI (21.7%), Sentinel-2 (20.7%), Landsat (12.3%), EnMAP (7.6%) and PRISMA (4.0%). The remaining records include VIIRS, Sentinel-3, Planet Tanager and GOES.

> Do not read this as a sensor ranking. Record counts depend on spatial resolution, detection limits, revisit, tasking, cloud/surface conditions, validation and data availability. They describe this MARS snapshot, not relative instrument quality.

TROPOMI context

The Sentinel-5P/TROPOMI Level-2 methane product provides column-averaged methane information with broad spatial coverage. UNEP IMEO technical documentation describes TROPOMI hotspot information as one input used to direct targeting and monitoring with higher-spatial-resolution satellites. This report does not reproduce that hotspot or retrieval methodology; it only handles the published downstream MARS records.

5. Sector and temporal composition

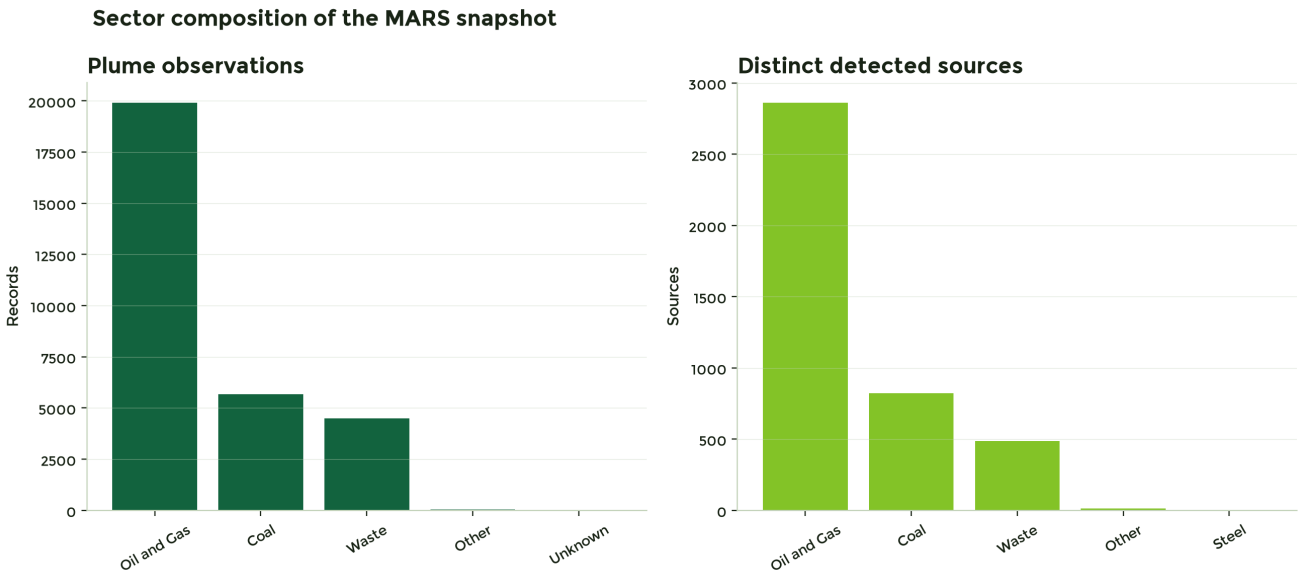


Figure 4. Sector composition of plume observations and distinct detected sources.

Oil and gas dominates both plume observations and detected sources in this snapshot. Coal and waste are the next largest groups. This is consistent with MARS' historical focus and should not be interpreted as a sector-neutral global methane inventory.

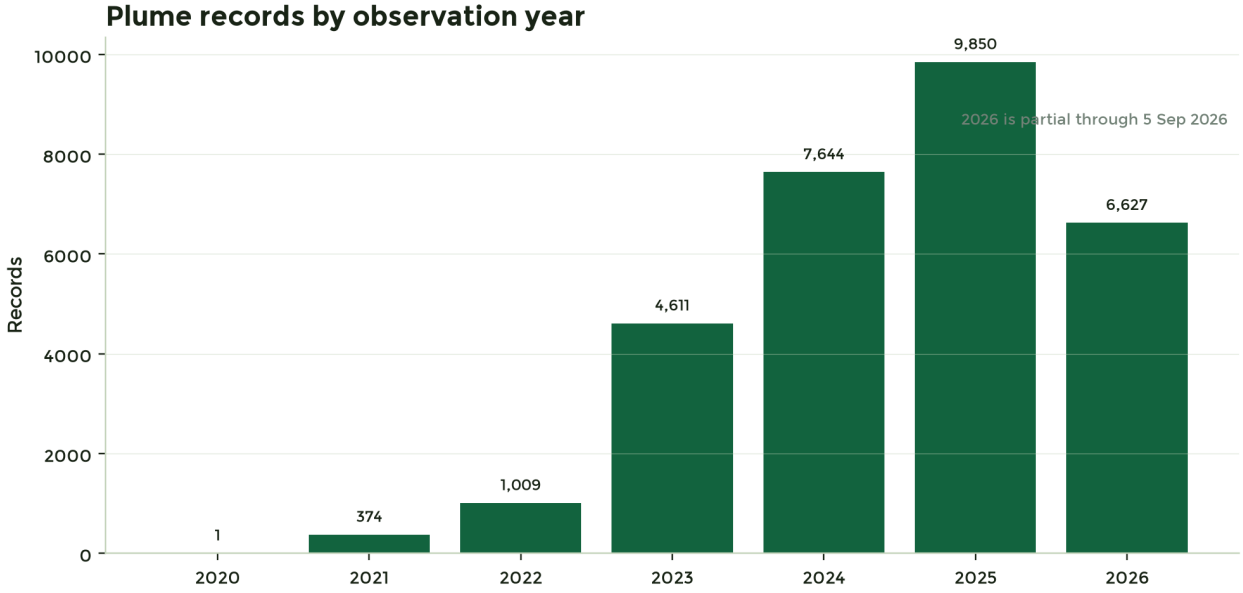


Figure 5. Plume records by observation year. The 2026 count is partial through 5 September.

The strong increase in records from 2023 onward reflects the evolving observing system, program operations, sensor availability, targeting, validation and public-data coverage. It is not a controlled time series of global methane emissions.

6. Published flux-rate fields

The public MARS plume export contains methane flux-rate estimates in kg/h for most records. After numeric validation, 29,058 records have a positive flux-rate value. These values are summarized exactly as published; the pipeline does not perform cross-sensor recalibration or uncertainty harmonization.

Statistic	Published positive flux rate
Valid positive records	29,058
25th percentile	1,393 kg/h
Median	3,027 kg/h
75th percentile	11,000 kg/h
95th percentile	73,000 kg/h

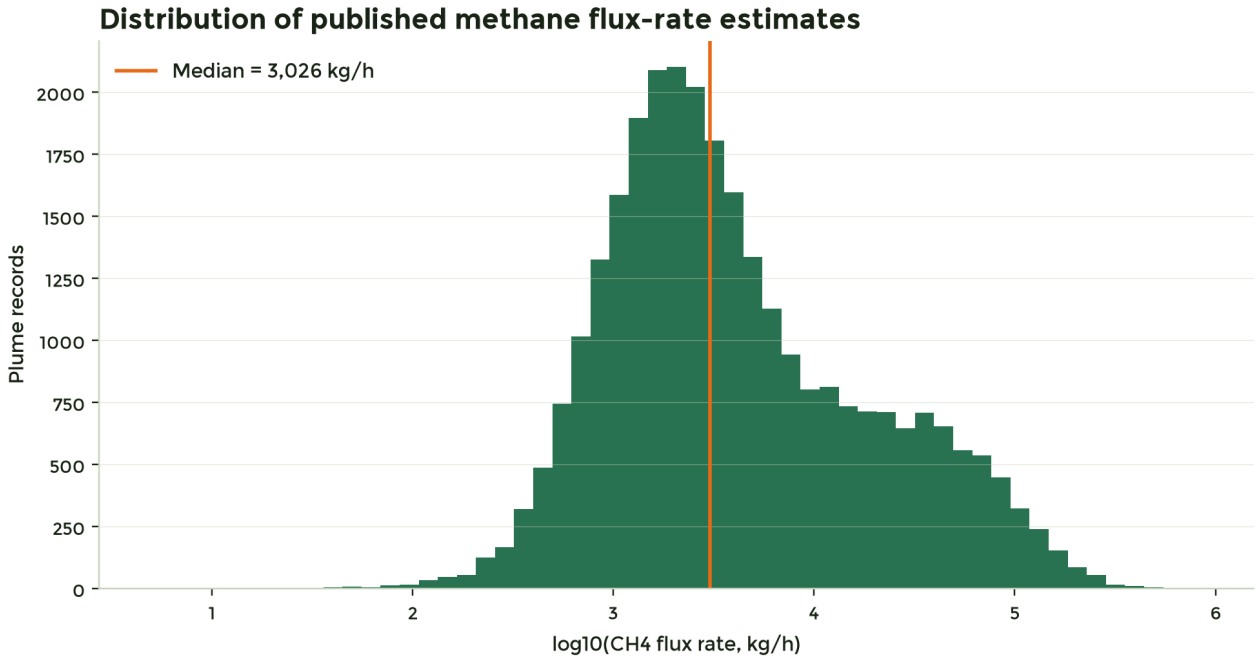


Figure 6. Distribution of positive published methane flux-rate estimates on a log10 x-axis.

Flux-rate comparisons across sensors require scientific context beyond this demonstration because instruments and quantification pathways differ. The figure is a data-field summary, not a cross-sensor performance analysis.

7. Spatial distribution

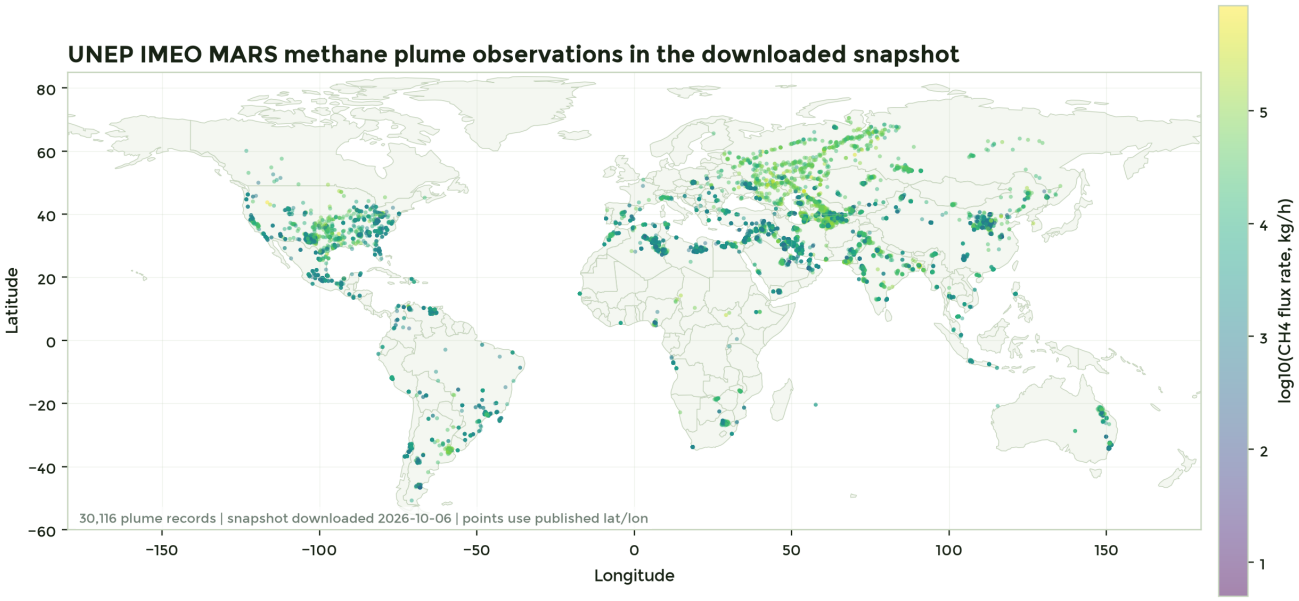


Figure 7. Global distribution of published plume locations. Positive flux-rate records are colored by log10(CH4 flux rate). Natural Earth boundaries are shown only for geographic context.

The snapshot spans 99 countries. The map represents where MARS has detected and published plume observations, not a complete global map of methane emissions. Coverage depends on instrument capability, observation opportunities, cloud and surface conditions, targeting, retrieval sensitivity, validation and program priorities.

The MARS data dictionary notes that Sentinel-5P source coordinates are approximate and automatically generated based on plume-mask and wind-direction information. The normalized point layer therefore preserves upstream coordinates without claiming greater spatial precision than the source product.

8. Reproducibility and operational path

The pipeline is designed to be rerun from the official data source rather than relying on a hand-curated local copy. The current repository includes download automation, provenance hashing, QA logic, tests, publication assets, a portable GeoPackage output, Docker packaging and a PostGIS loading path.

Rebuild sequence

```
python src\download_data.py
python -m pytest -q
python src\run_pipeline.py
python src\publication_assets.py
python src\build_publication_report.py
```

Executed evidence

Component	Observed result
Automated tests	2/2 passed
Normalized GeoPackage	30,116 records, EPSG:4326
Inspection sample	1,000 GeoJSON records
Machine summaries	CSV + JSON
PostGIS	Schema + GIST/time/satellite indexes + UPSERT loader
Containerization	Dockerfile included

GeoPackage is the default executed artifact so the demonstration can be reproduced without provisioning a database server. The PostGIS path remains explicit for migration to an operational service-backed environment.

9. Limitations, licensing and responsible interpretation

Limitation	Interpretation
No atmospheric retrieval	The project starts from already processed MARS detections.
Snapshot dependence	Results are tied to the 6 October 2026 archive and SHA-256 hash.
No completeness claim	Public MARS observations are not a complete census of methane emissions.
No sensor ranking	Observation counts reflect dataset composition and observing strategy.
No trend inference	Annual counts are not a controlled global methane-emission time series.
No cross-sensor harmonization	Published flux rates are not recalibrated across retrieval approaches.
Point normalization	Missing plume polygons are not reconstructed; published lat/lon are used.

Licensing and attribution

The UNEP Eye on Methane MARS dataset is shared under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0). Derived figures, tables, the lightweight sample and this report are distributed under the same license. The independently authored pipeline source code is distributed under the MIT License.

Attribution statement. Contains data derived from UNEP International Methane Emissions Observatory (IMEO), Eye on Methane / Methane Alert and Response System (MARS), downloaded 6 October 2026. Data were normalized, quality-flagged, aggregated and visualized for this independent technical demonstration. Original MARS data are licensed CC BY-NC-SA 4.0. This work is not produced by or endorsed by UNEP or IMEO.

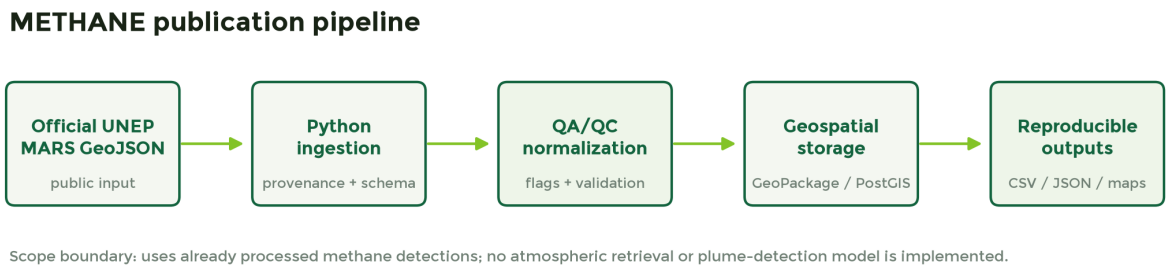
Natural Earth vector boundaries used for cartographic context are public-domain data. The intended publication model is a GitHub software repository plus a separately citable Zenodo technical report with its own DOI.

10. Conclusions

A compact downstream workflow is sufficient to demonstrate capabilities that matter for operational environmental Earth-observation systems: provenance, schema discipline, geospatial normalization, deterministic validation, explicit quality flags, reproducible summaries, spatial persistence and transparent limitations.

The main contribution is not a new methane retrieval algorithm. It is a defensible data-engineering layer around a real public methane EO product. The workflow receives an official MARS snapshot, validates its structure, preserves source provenance, creates a portable spatial representation, exposes a PostGIS loading path, and produces machine-readable and illustrated outputs without overstating the scientific scope.

This boundary is deliberate: it demonstrates how established remote-sensing and geospatial engineering practices can be transferred to methane data while preserving the distinction between data engineering and atmospheric retrieval science.



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Code repository: https://github.com/marcel-innocentini/methane-imeo-data-engineering
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